Ch3. Applications of Differentiation

§3.1 Maximum and Minimum Values

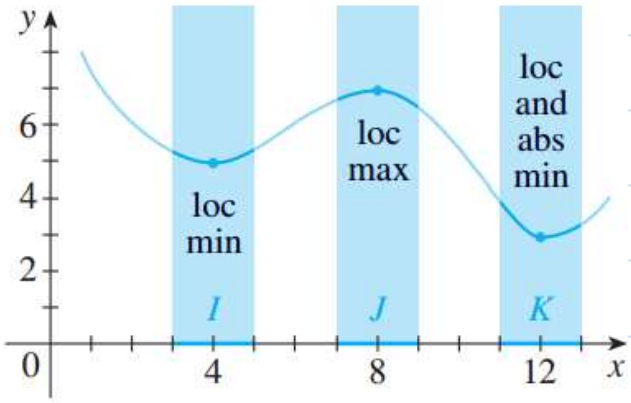
1 DEFINITION Let c be a number in the domain D of a function f. Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(c) \geqslant f(x)$ for all x in D.
- **absolute minimum** value of f on D if $f(c) \leqslant f(x)$ for all x in D.

Extreme values of f : the absolute / global maximum and minimum values of f

2 DEFINITION The number $f(c)$ is a

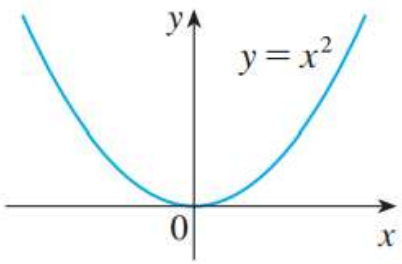
- **local maximum** value of f if $f(c) \geqslant f(x)$ when x is near c.
- **local minimum** value of f if $f(c) \leqslant f(x)$ when x is near c.



When x is near c

When x is in some open interval containing c

Example Let $f(x) = x^2$.



Example Let $f(x)=3x^4-16x^3+18x^2 \ \ (-1 \le x \le 4)$. Determine the local/absolute extreme values of f.

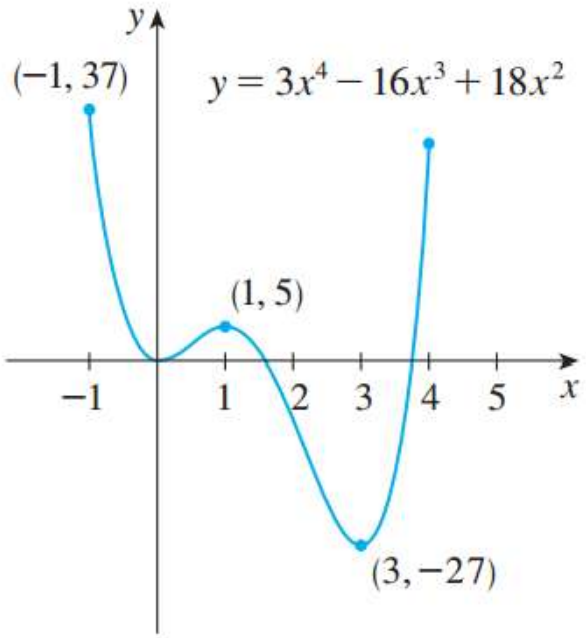
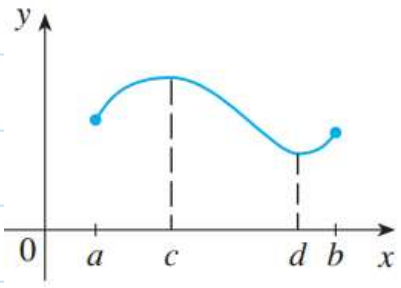
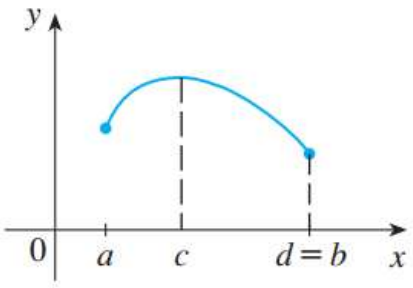
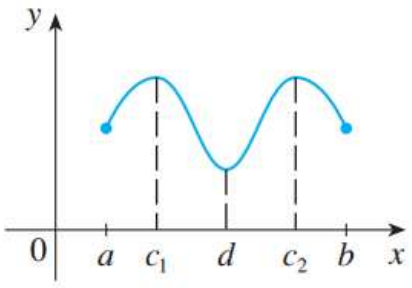


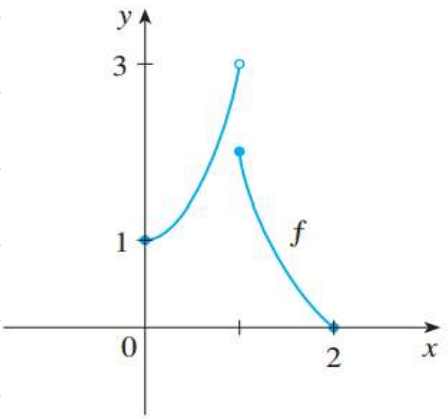
FIGURE 6

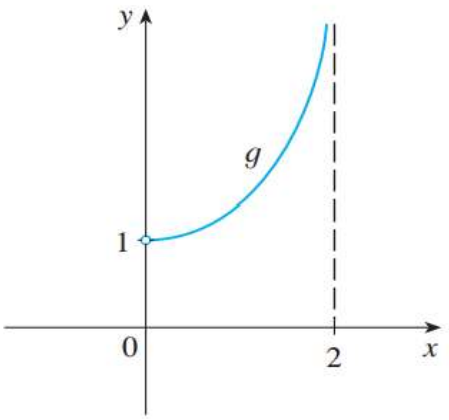
3 THE EXTREME VALUE THEOREM If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.











4 FERMAT'S THEOREM If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.

Note The converse of Fermat's Theorem is false in general.

6 DEFINITION A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Example Find the critical numbers of $f(x) = x^{3/5}(4-x)$.

[revisited]

7 If f has a local maximum or minimum at c, then c is a critical number of f.

THE CLOSED INTERVAL METHOD To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b).
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example Find the absolute maximum and minimum values of the function

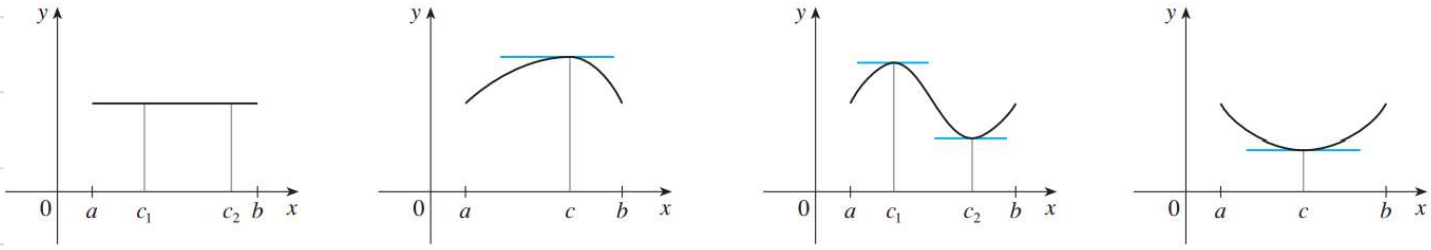
$$f(x) = x^3 - 3x^2 + 1 \qquad -\frac{1}{2} \le x \le 4$$

§3.2 The Mean Value Theorem

ROLLE'S THEOREM Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.



Example Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root.

THE MEAN VALUE THEOREM Let f be a function that satisfies the following hypotheses:

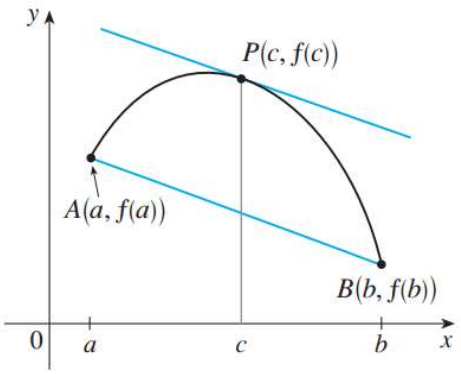
1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b).

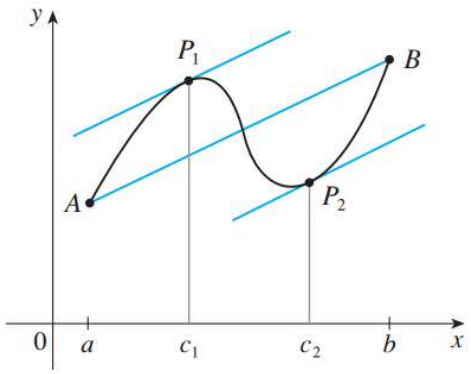
Then there is a number c in (a, b) such that

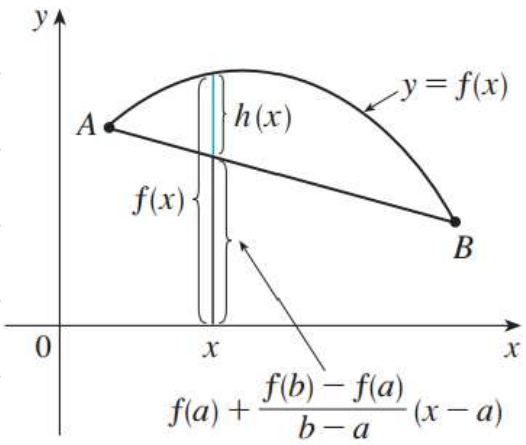
1 $$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

2 $$f(b) - f(a) = f'(c)(b - a)$$





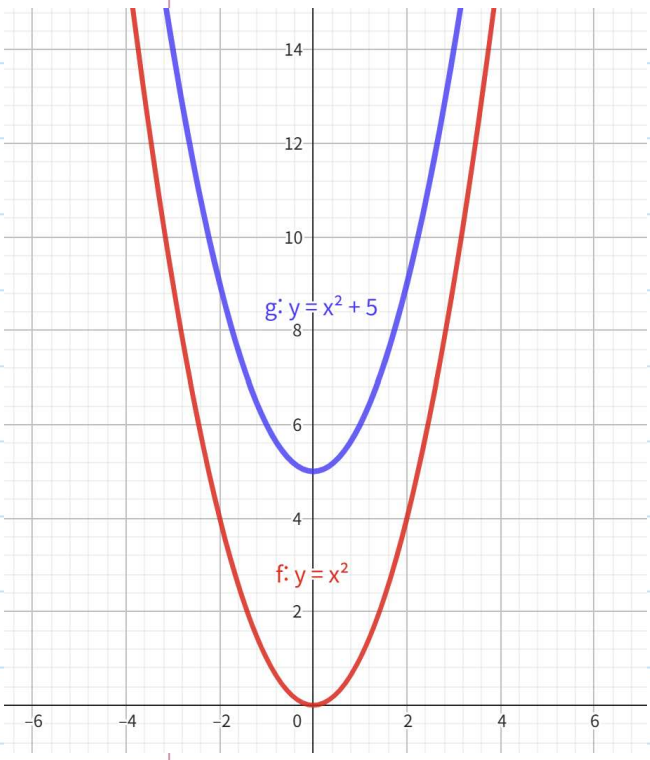


Example Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x. How large can $f(2)$ possibly be?

5 THEOREM If $f'(x) = 0$ for all x in an interval (a, b), then f is constant on (a, b).

7 COROLLARY If $f'(x) = g'(x)$ for all x in an interval (a, b), then $f - g$ is constant on (a, b); that is, $f(x) = g(x) + c$ where c is a constant.

Example $f(x) = x^2$ and $g(x) = x^2 + 5$.



Example $f(x) = \dfrac{x}{|x|}$

Ch3. Applications of Differentiation

§3.3 Derivatives and the Shapes of Graphs

Theorem [Increasing / Decreasing Test]

(a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.

(b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

why) $\dfrac{f(x_2 - x_1)}{x_2 - x_1} = f'(c)$ for some $c \in (x_1, x_2)$

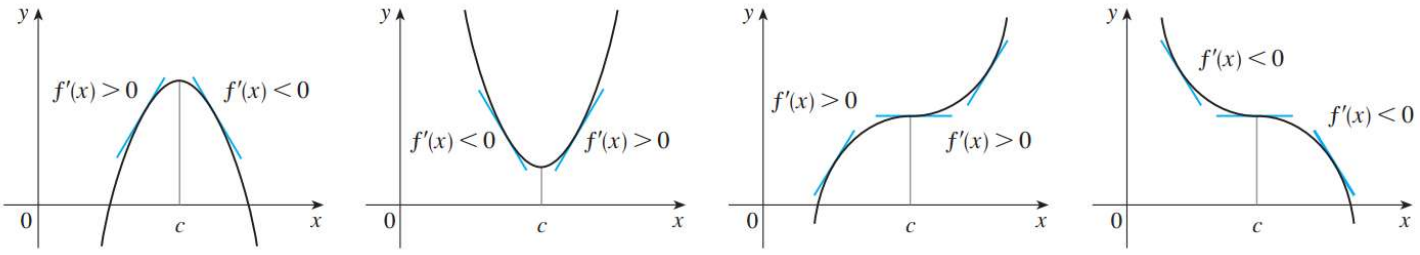
Theorem [The First Derivative Test]

Suppose that c is a critical number of a continuous function f.

(a) If f' changes from positive to negative at c, then f has a local maximum at c.

(b) If f' changes from negative to positive at c, then f has a local minimum at c.

(c) If f' does not change sign at c, then f has no local maximum or minimum at c.



(a) Local maximum (b) Local minimum (c) No maximum or minimum (d) No maximum or minimum

FIGURE 3

Example Let $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$.

(a) Find where the function f is increasing and where it is decreasing.

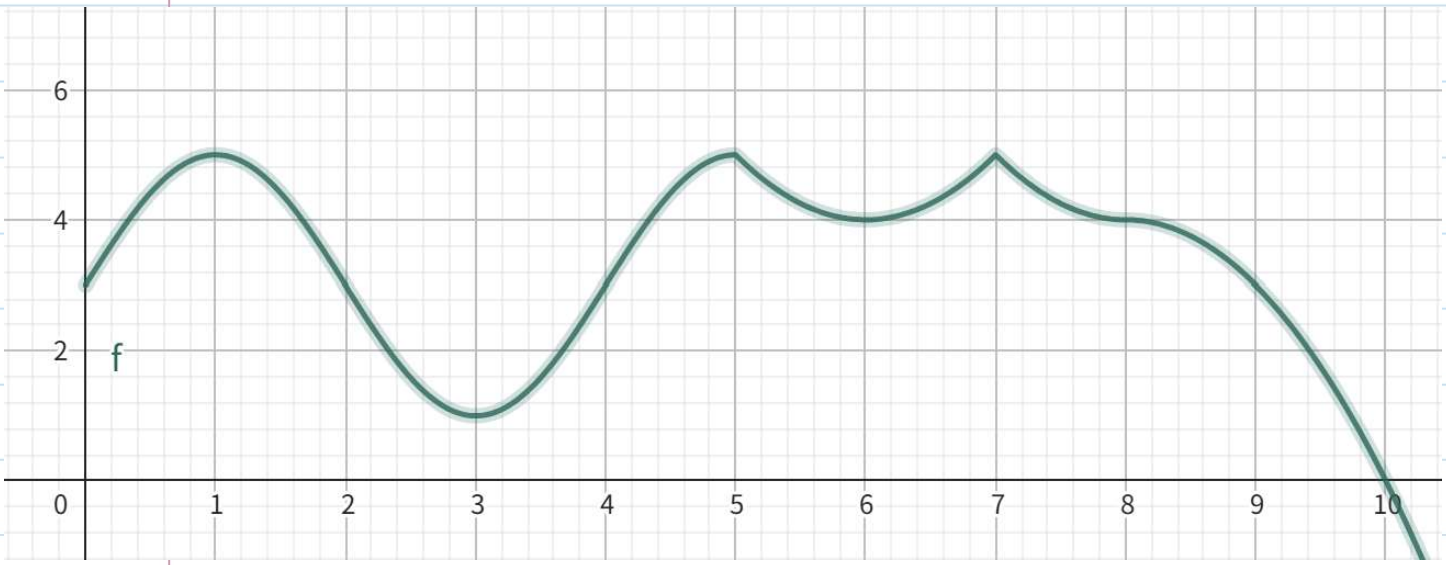
(b) Find the local minimum and maximum values of the function f.

Example Find the local minimum and maximum values of the function

$$g(x) = x + 2\sin x,\ 0 \le x \le 2\pi.$$

Definition If the graph of f lies above all of its tangents on the interval I, then it is called **concave upward** on I.
If the graph of f lies below all of its tangents on the interval I, then it is called **concave downward** on I.

Definition A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P.



$$g(x) = \begin{cases} 2\sin(x\pi/2) + 3 & \text{if } 0 \le x \le 5 \\ (x-6)^2 + 4 & \text{if } 5 \le x \le 7 \\ (x-8)^2 + 4 & \text{if } 7 \le x \le 8 \\ -(x-8)^2 + 4 & \text{if } 8 < x \end{cases}$$

Theorem [Concavity Test]

(a) If $f''(x) > 0$ for all $x \in I$, then the graph of f is concave upward on I.

(b) If $f''(x) < 0$ for all $x \in I$, then the graph of f is concave downward on I.

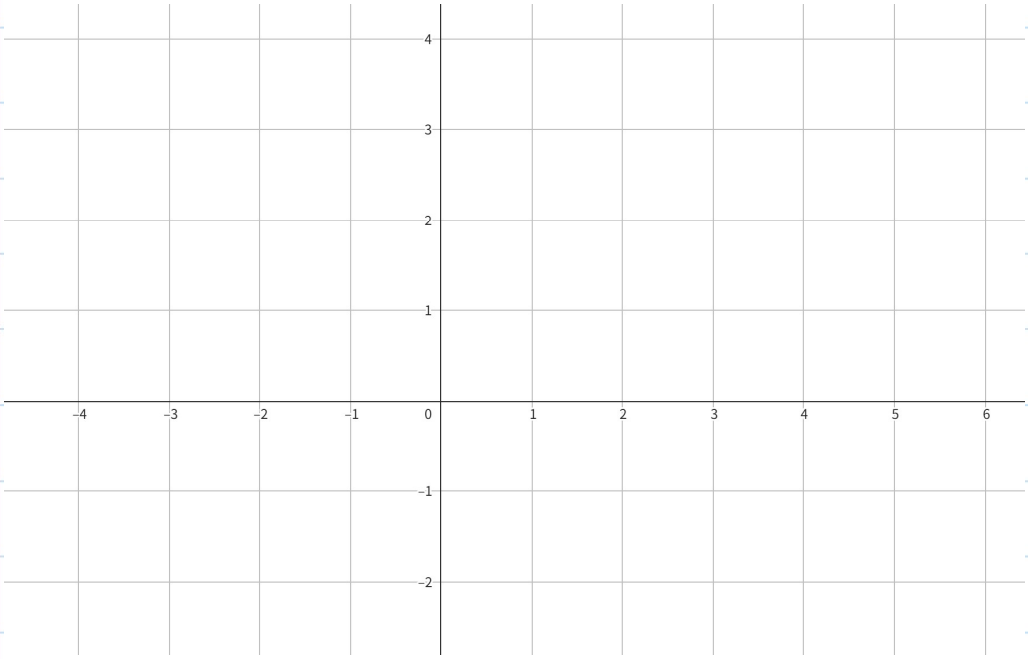
$$g(x) = \begin{cases} 2\sin(x\pi/2) + 3 & \text{if } 0 \le x \le 5 \\ (x-6)^2 + 4 & \text{if } 5 \le x \le 7 \\ (x-8)^2 + 4 & \text{if } 7 \le x \le 8 \\ -(x-8)^2 + 4 & \text{if } 8 < x \end{cases}$$

Example Sketch a possible graph of a function f that satisfies the following conditions :

(i) $f'(x) > 0$ on $(-\infty, 1)$, $f'(x) < 0$ on $(1, \infty)$

(ii) $f''(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$, $f''(x) < 0$ on $(-2, 2)$

(iii) $\lim_{x \to \infty} f(x) = -2$, $\lim_{x \to \infty} f(x) = 0$



Theorem [The Second Derivative Test]

Suppose f'' is continuous near c.

(a) If $f'(c)=0$ and $f''(c)>0$, then f has a local minimum at c.

(b) If $f'(c)=0$ and $f''(c)<0$, then f has a local maximum at c.

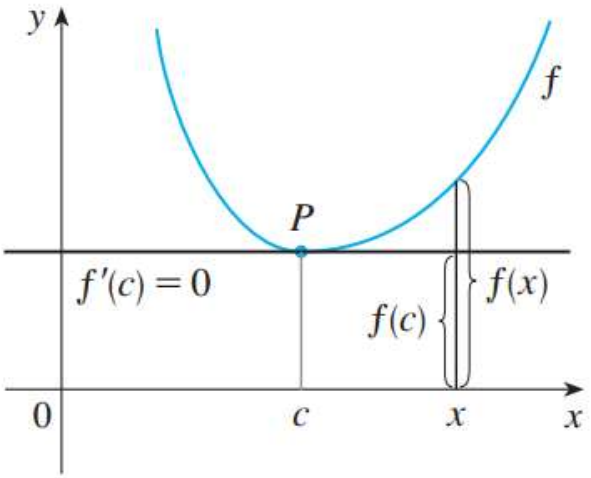


FIGURE 9

$f''(c)>0$, f is concave upward

Example Discuss the curve $y=f(x)=x^4-4x^3$ with respect to concavity, points of inflection, and local maxima and minima. Use this information to sketch the curve.

§3.4 Curve Sketching

[Guidelines for Sketching a Curve]

Given a function f, use the following steps to sketch a graph of f :

A. Domain :

Determine the domain of f , the set of values of x for which $f(x)$ is defined.

B. Intercepts :

Locate the x- and y-intercepts.

(You can omit this step if the equation is difficult to solve.)

C. Symmetry :

(i) even function : $f(-x)=f(x)$ for every $x\in D=\mathrm{dom}(f)$

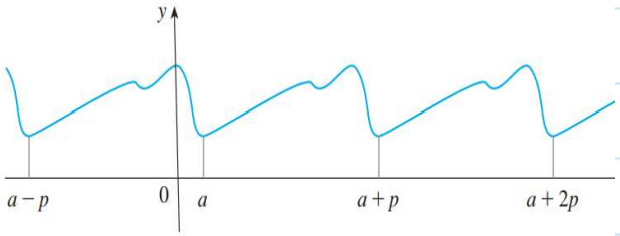
f is symmetric about the y-axis.

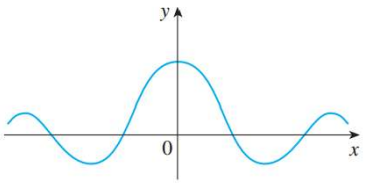
(ii) odd function : $f(x)=-f(-x)$ for every $x\in D=\mathrm{dom}(f)$

f is symmetric about the origin, this means rotate 180° about the origin, you can get the same graph.

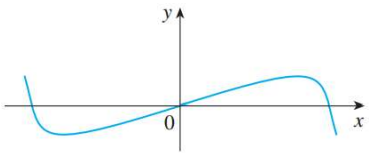
(iii) periodic function : $f(x+p)=f(x)$ for every $x\in D=\mathrm{dom}(f)$

f is periodic with the **period** p, when p is the smallest such number.





(a) Even function: reflectional symmetry



(b) Odd function: rotational symmetry

D. Asymptotes

(i) Horizontal Asymptotes :

Evaluate $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ to determine the end behavior. If either of these limits is a finite number L, then $y=L$ is a horizontal asymptote.

(ii) Vertical Asymptotes :

Determine whether f has any vertical asymptotes.

If at least one of the following is true, then $x=a$ is a vertical asymptote:

$$\lim_{x\to a^{\pm}} f(x)=\infty \qquad\qquad \lim_{x\to a^{\pm}} f(x)=-\infty$$

E. Intervals of Increase or Decrease :

Compute $f'(x)$ and find the invervals on which f' is positive and the intervals on which f' is negative.

F. Local Maximum and Minimum Values :

Find the critical numbers of f. Then use the First Derivative Test or the Second Derivative Test to find local extrema.

G. Concavity and Points of Inflection :

Compute $f''(x)$ and use the Concavity Test to determine the intervals where f is concave upward and where f is concave downward.

Inflection points occur where the direction of concavity changes.

H. Sketch the curve :

Sketch the asymptotes as dashed lines. Plot the intercepts, maximum and minimum points, and inflection points. TThen make the curve pass through these points, rising and falling according to E, with concavity according to G, and approaching the asymptotes.

If additional accuracy is desired near any point, you can compute the value of the derivative there. The tangent indicates the direction in which the curve proceeds.

Example Use the guidelines to sketch the curve $y = \dfrac{2x^2}{x^2-1}$.

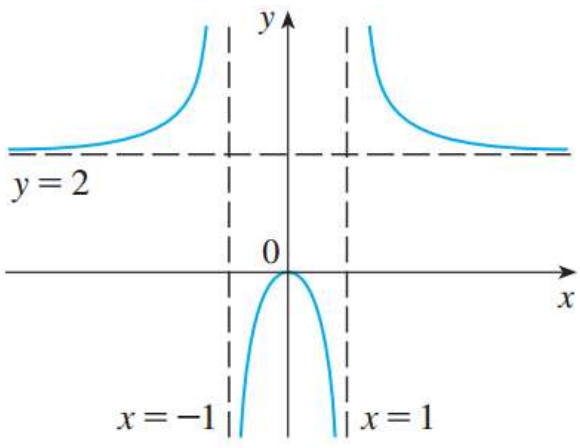


FIGURE 4
Finished sketch of $y = \dfrac{2x^2}{x^2 - 1}$

§3.5 Optimization Problems

[Steps in Solving Optimization Problems]

1. **Understand the Problem** The first step is to read the problem carefully until it is clearly understood. Ask yourself: What is the unknown? What are the given quantities? What are the given conditions?
2. **Draw a Diagram** In most problems it is useful to draw a diagram and identify the given and required quantities on the diagram.
3. **Introduce Notation** Assign a symbol to the quantity that is to be maximized or minimized (let's call it Q for now). Also select symbols $(a, b, c, \ldots, x, y)$ for other unknown quantities and label the diagram with these symbols. It may help to use initials as suggestive symbols—for example, A for area, h for height, t for time.
4. Express Q in terms of some of the other symbols from Step 3.
5. If Q has been expressed as a function of more than one variable in Step 4, use the given information to find relationships (in the form of equations) among these variables. Then use these equations to eliminate all but one of the variables in the expression for Q. Thus Q will be expressed as a function of *one* variable x, say, $Q = f(x)$. Write the domain of this function.
6. Use the methods of Sections 3.1 and 3.3 to find the *absolute* maximum or minimum value of f. In particular, if the domain of f is a closed interval, then the Closed Interval Method in Section 3.1 can be used.

Example A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Example Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

Example Find the area of the largest rectangle that can be inscribed in a semicircle of radius r.

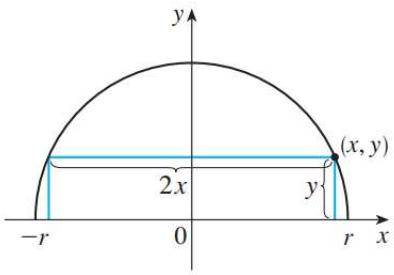


FIGURE 9

Ch3. Applications of Differentiation

§3.7 Antiderivatives

Definition A function F is called an **antidervative** of f on an interval I if $F'(x)=f(x)$ for all $x\in I$.

examples $f(x)=x^2$.

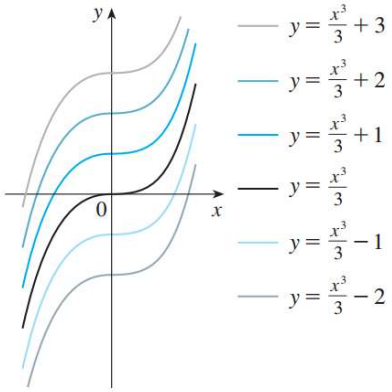


FIGURE 1
Members of the family of antiderivatives of $f(x)=x^2$

Theorem If F is an antiderivative of f on an interval I, then the most general antiderivative of f on I is

$$F(x)+C$$

where C is an arbitrary constant.

Example Find f if $f'(x)=x\sqrt{x}$ and $f(1)=2$.

Example Find f if $f''(x)=12x^2+6x-4$, $f(0)=4$ and $f(1)=1$.